

Project Report: Predicting Firm Success

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1. Introduction

Project Objective & Implementation: This project was launched to develop a predictive algorithm designed to identify firms with high growth potential and low investment risk. By refining our success-prediction models using 2013 data, we have created a tool that allows for more data-driven capital allocation. Starting next month, the finance department will integrate these algorithms into their workflow to optimize their investment strategies.

Data & Methodology: The analysis utilized a cross-sectional dataset of 15,231 European firms. A key focus was placed on "fast-growth" companies, which are defined as those with a sales increase exceeding 0.3% between 2012 and 2013 – which was achieved by 3,473 firms in total. We evaluated over 200 predictive variables, ranging from financial indicators (assets, liabilities, and profits) to qualitative factors (management profiles, firm age, and location).

Model Selection & Industry Insights: To ensure accuracy, we tested multiple machine learning models and selected the top performer based on two critical metrics: the lowest overall prediction error and the minimized financial loss (in USD) resulting from wrong predictions. Furthermore, we conducted a comparative analysis between the **manufacturing** and **services** sectors. This breakdown allows us to determine if specific models perform better within certain industries and assess their robustness across domains.

2. Data and Methodology

We defined **"firm success"** as **year-on-year sales growth exceeding 0.3%** as the **key performance indicator**. This was selected because it is a straightforward measure that is not directly affected by investment decisions, national tax laws, or internal accounting decisions. A one-year horizon was chosen both to align with standard investment cycles as well as to take advantage of higher data quality in the respective year.

We evaluated our models based on:

- **Area Under the Curve (AUC):** This measures the model's ability to distinguish between successful and unsuccessful firms. An AUC of 1 is a perfect score, while 0.5 is no better than a coin flip.
- **Prediction Error (RMSE/Brier):** Goal is to minimize the error in our predictions.
- **Expected Loss:** A metric that weighs the financial impact of missed opportunities versus bad investments. We determined that wrongly investing in a slow-growth firm (**False Positive**) should be twice as costly as missing an investing opportunity of high-growing one (**False Negative**). We adjusted our "decision threshold" to reflect this priority.

After testing several standard regressions (**Logit** and **LASSO logit**) and advanced machine learning (**Gradient Boosting** and **Random Forest**), we compared the best performing models of each model type in terms of prediction error and AUC. According to these metrics, Gradient Boosting and Random Forest outperformed the regression models.

Comparison of the four best performing models

Model	RMSE/Brier	AUC
Logit_Model_X4	0.3837	0.758
LASSO Logit	0.3817	0.761
Gradient Boosting	0.3753	0.781
Random Forest	0.3770	0.774

After narrowing our focus to four primary models, we calculated the expected average loss, based on the weight attributed to False Positives (2/3) and False Negatives (1/3) at each of the model's optimal thresholds. A threshold is a "decision point": if the probability of fast growth is above this number, we classify it as success, if below, we do not. We can conclude that the Random Forest model results in the lowest loss of \$63,6k (per firm) at a threshold of 0.60.

Comparison of thresholds and expected loss

Model	Chosen threshold	Expected loss (USD)
Logit X4	0.650	64,665
Lasso Logit	0.625	64,665
Random Forest	0.600	63,559
Gradient Boosting	0.675	64,223

3. Business Impact of Random Forest Model

The following model metrics aim to give a more in-depth insight to the performance of the chosen Random Forest model.

Model Metrics

Metric	Value	Description
Accuracy	80.32	Overall correctness of predictions
Precision	76.57	Reliability of "Buy" signals
Recall	21.37	Ability to identify all profitable opportunities
Specificity	98.04	Ability to correctly identify and avoid losses
F1-Score	0.3342	Harmonic mean of Precision and Recall

- **False Positives:** The model identifies 175 companies for investment, from which 41 were "traps" (False Positives). From a capital preservation standpoint, this means 3 out of every 4 dollars invested are hitting the "Fast Growth" target; Precision is 76.5%.
- **False Negatives:** The model is highly conservative, missing 493 high-growth opportunities. While the cumulative "Missed Profit" is high (\$147.9M), this aligns with the business mandate to prioritize safety over volume.
- **Risk Mitigation:** It effectively filters out over 98% of the "Other/Default" companies (2,046 out of 2,087).
- The model recommends 175 investments. Thus, the model provides a curated list for potential investments while maintaining a high safety margin.

4. Model Performance across Industries

In Addition to the calculations above, we carried out an analysis to compare the performance of the Random Forest model between the **construction** and **service sector**. Overall, the statistical performance on the two subsets is quite similar to the full holdout set, but there is one significant difference: the model seems to be much better in catching fast growing firms in the services sector (25.5% recall) than in manufacturing (11% recall). This could be due to the fact that there was more training data from services, so the model was able to make more confident predictions about the growth of service sector companies.

Model Metrics – Comparing Manufacturing and Services

Metric	Manufacturing	Services	Description
Accuracy	80.12%	80.42%	Overall correctness of predictions
Precision	76.92%	76.51%	Reliability of "Buy" signals
Recall	11.05%	25.56%	Ability to identify all profitable opportunities
Specificity	99.09%	97.55%	Ability to correctly identify and avoid losses
F1-Score	0.1932	0.3832	Harmonic mean of Precision and Recall

5. Discussion

After comprehensive testing of multiple machine learning frameworks, the Random Forest model was selected as the primary basis for future investment decisions. By weighting cost of “incorrect investments” (False Positives) twice as heavy as “missed opportunities” (False Negatives), our chosen model achieves the lowest expected loss of \$63,559 per firm, compared to other candidates. Performance remains reliable in both the manufacturing and service sectors, but the model performs significantly better in the latter. The low recall in the manufacturing sector creates substantial opportunity cost and business loss. Future iterations should focus on selecting a larger sample of firms in that industry and explore domain-specific characteristics. Despite this, it is highly recommended to apply the model for investment decisions across all industries. It operates with a conservative bias, with the aim of shielding the portfolio from volatility and maximizing the cost reduction associated with high-risk and high-gain investments.